

#1 Determine en $\mathbb{C}$ la factorización completa del polinomio

(3 Pts)

$$P(z) = 9z^5 + 27z^4 + 28z^3 + 12z^2 + 3z + 1$$

si se sabe que $z = i/3$ es un cero $P(z)$. Debe mostrar los procedimientos detallados que le permitieron obtener los demás ceros de $P(z)$.

$$9z^5 + 27z^4 + 28z^3 + 12z^2 + 3z + 1$$

$$\begin{array}{r|l} 9 & 27 & 28 & 12 & 3 & 1 \\ & 3i & 9i-1 & 9i-3 & -3+3i & -1 \\ \hline 9 & 27+3i & 27+9i & 9i+9 & 3i & 0 \end{array} \quad \begin{array}{l} \frac{i}{3} \\ (x - \frac{i}{3}) \end{array}$$

$\frac{i}{3} \cdot 9 = 3i$

$$\begin{array}{r|l} 9 & 27+3i & 27+9i & 9i+9 & 3i \\ & -3i & -9i & -9i & -3i \\ \hline 9 & 27 & 27 & 9 & 0 \end{array} \quad \begin{array}{l} \frac{i}{3} (27+3i) \\ 9i+9 \\ (x - \frac{i}{3}) \end{array}$$

$\frac{i}{3} (27+9i)$
 $9i-3$

$$9z^3 + 27z^2 + 27z + 9$$

$$9(z^3 + 3z^2 + 3z + 1)$$

$$\begin{array}{r|l} 1 & 3 & 3 & 1 \\ & -1 & -2 & -1 \\ \hline 1 & 2 & 1 & 0 \end{array} \quad \begin{array}{l} -1 \\ \pm 1 \end{array} \quad \begin{array}{l} \frac{i}{3} \cdot 3i \\ i^2 = -1 \end{array} \quad \begin{array}{l} -3+3i \\ \end{array}$$

$(x - (-1))$
 $(x + 1)$

$$z^2 + 2z + 1$$

$$\begin{array}{l} z \quad \quad \quad 1 = z \\ z \quad \quad \quad 1 = z \\ \hline 2z \end{array} \quad x = z$$

$$(x+1)(x+1) = (x+1)^2 \quad (x+1) = (x+1)^3$$

$$\boxed{9(x - \frac{i}{3})(x + \frac{i}{3})(x + 1)^3}$$

#2 Usando el teorema de DeMoivre calcule y exprese en forma polar el resultado de (4 Pts)

$$\frac{\overset{z1}{(-1+i\sqrt{3})}^6}{16i^{15} \overset{z2}{22}} r \operatorname{cis}(\theta)$$

$$z1 = -1 + i\sqrt{3}$$

$$a = -1 \quad b = \sqrt{3} \quad -a + b \rightarrow +\pi$$

$$r = \sqrt{(-1)^2 + (\sqrt{3})^2} = 2$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{3}}{-1}\right) + \pi = \frac{2\pi}{3} \quad i^0 = 1 \quad i^2 = -1$$

$$z1 = 2 \operatorname{cis}\left(\frac{2\pi}{3}\right) \quad i^1 = i \quad i^3 = -i$$

$$z2 = 16i^{15} \quad i^{15} \rightarrow h - \operatorname{int}\left(\frac{h}{4}\right)$$

$$= -16i + 0 \quad 15 - \operatorname{int}\left(\frac{15}{4}\right)$$

$$b \quad a \quad 15 - 4 \cdot 3 = 3$$

$$a = 0 \quad b < 0 \rightarrow \theta = -\frac{\pi}{2}$$

$$r = \sqrt{0^2 + (-16)^2} = 16$$

$$z2 = 16 \operatorname{cis}\left(-\frac{\pi}{2}\right)$$

$$\frac{\overset{z1}{(-1+i\sqrt{3})}^6}{16i^{15} \overset{z2}{22}}$$

$$\frac{(2 \operatorname{cis}\left(\frac{2\pi}{3}\right))^6}{16 \operatorname{cis}\left(-\frac{\pi}{2}\right)}$$

$$(r \operatorname{cis}(\theta))^h$$

$$r^h \operatorname{cis}(\theta \cdot h)$$

$$\frac{2^6 \cdot \operatorname{cis}\left(\frac{2\pi}{3} \cdot 6\right)}{16 \operatorname{cis}\left(-\frac{\pi}{2}\right)}$$

$$\frac{2^6}{2^4} \operatorname{cis}\left(\frac{12\pi}{3} - -\frac{\pi}{2}\right)$$

$$\boxed{16 \cdot \operatorname{cis}\left(\frac{9\pi}{2}\right)}$$

#3 Sea $z = 1 - e^{\frac{\pi}{2}i}$.

$$\text{cis } \theta = re^{i\theta}$$

a) Compruebe que $z = 1 - i$.

$$1 - e^{\frac{\pi}{2}i}$$

$$1 - \left[\text{cis} \left(\frac{\pi}{2} \right) \right]$$

$$1 - \left[\cos \left(\frac{\pi}{2} \right) + \text{sen} \left(\frac{\pi}{2} \right) i \right]$$

$$1 - [0 + 1i]$$

$$\boxed{1 - i}$$

b) Calcule $\text{Ln}(z)$. $z = 1 - i$

$$\ln(z) = \ln(1 - i)$$

$$z = 1 - i \quad a = 1 \quad b = -1$$

$$r = \sqrt{1 + (-1)^2} = \sqrt{2} \quad \theta = \arctan \left(\frac{-1}{1} \right) = -\frac{\pi}{4}$$

$$z = \sqrt{2} \cdot e^{-i\frac{\pi}{4}}$$

$$\ln(z) = \ln \left(\sqrt{2} \cdot e^{-\frac{\pi}{4}i} \right)$$

$$\ln(a \cdot b)$$

$$= \ln(a) + \ln(b)$$

$$= \ln(\sqrt{2}) + \ln \left(e^{-\frac{\pi}{4}i} \right)$$

$$\boxed{\ln(\sqrt{2}) - \frac{\pi}{4}i}$$

#4 Considere el número $z \in \mathbb{C}$ tal que:

$$|z - 3 - 4i| = 5 \quad \text{y} \quad \text{Arg}(\bar{z} + i) = -\frac{\pi}{2}$$

Determine todos los posibles valores de $z \in \mathbb{C}$ que cumplan ambas condiciones. (4 Pts)

$$z = a + bi$$

$$|z - 3 - 4i| = 5$$

$$|a + bi - 3 - 4i| = 5$$

$$|(a-3) + (b-4)i| = 5$$

$$(a-3)^2 + (b-4)^2 = 25 \quad \text{Can't be solve yet}$$

$$\arg(\bar{z} + i) = -\frac{\pi}{2}$$

$$a = 0 \quad b < 0$$

$$\overline{a + bi} + i = -\frac{\pi}{2}$$

$$a - bi + i = -\frac{\pi}{2}$$

$$a + (1-b)$$

↓

$$a = 0$$

↓

$$1 - b < 0$$

$$1 - b < 0$$

$$-b < -1$$

$$b > 1$$

$$(a-3)^2 + (b-4)^2 = 25 \quad a = 0$$

$$9 + (b-4)^2 = 25$$

$$(b-4)^2 = 16$$

$$b-4 = \pm 4$$

$$b = 0 \quad \times \quad b > 1$$

$$b = 8 \quad \checkmark \quad b > 1$$

$$\boxed{z = 8i}$$

Dependiendo de esta tabla lo de $\tan^{-1}(\frac{b}{a})$ hay sumar o restar π

Cuadrante	a	b	Rango	Que hago?
I	+	+	Ya esta	NADA
II	-	+	Esta Fuera	$+\pi$ al θ
III	-	-	Esta Fuera	$-\pi$ al θ
IV	+	-	Ya esta	NADA

$$\theta = \frac{\pi}{2} \quad \text{si } a = 0 \quad \text{y } b > 0 \quad z = 3i$$

$$\theta = -\frac{\pi}{2} \quad \text{si } a = 0 \quad \text{y } b < 0 \quad z = -3i$$

$$\theta = \left[\tan^{-1}\left(\frac{b}{a}\right) \right] \pm \pi$$

#5 Sean A, B, C y Q matrices reales tales que $AB + 2C - Q = 0$, con

$$A = \begin{pmatrix} a & 0 & 2 \\ 3 & 1 & b \end{pmatrix}, \quad B = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 3 & 1 \\ a & 2 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 3a & 1 & 0 \\ -1 & -b & -4 \end{pmatrix} \begin{matrix} 3 \\ 2 \times 3 \end{matrix}$$

a) Determine qué tamaño tiene la matriz Q . Justifique.

$$AB + 2C - Q = 0$$

$$Q = AB + 2C$$

$$\begin{pmatrix} a & 0 & 2 \\ 3 & 1 & b \end{pmatrix} \times \begin{pmatrix} 2 & -1 & 3 \\ 0 & 3 & 1 \\ a & 2 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2a+0+2a & -a+0+7 & 3a+0+7 \\ 6+0+ab & -3+3+2b & 9+1+2b \end{pmatrix}$$

$$\begin{pmatrix} 4a & 7-a & 3a+7 \\ 6+ab & 2b & 10+2b \end{pmatrix}$$

$$C = \begin{pmatrix} 3a & 1 & 0 \\ -1 & -b & -4 \end{pmatrix}$$

$$2C = \begin{pmatrix} 6a & 2 & 0 \\ -2 & -2b & -8 \end{pmatrix}$$

$$\begin{pmatrix} 4a & 7-a & 3a+7 \\ 6+ab & 2b & 10+2b \end{pmatrix} + \begin{pmatrix} 6a & 2 & 0 \\ -2 & -2b & -8 \end{pmatrix}$$

$$Q = \begin{pmatrix} 10a & 6-a & 3a+7 \\ 7+ab & 0 & 2+2b \end{pmatrix}$$

Q debe ser 2×3

b) Calcule únicamente el valor de las entradas $\langle Q \rangle_{11}$ y $\langle Q \rangle_{22}$.

$$Q = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{pmatrix} 10a & 6-a & 3a+7 \\ 7+ab & 0 & 2+2b \end{pmatrix} \end{matrix}$$

$$Q_{11} = 10a \quad Q_{22} = 0$$

$$X^T = X$$

#6 Considere la ecuación matricial

$$AX + (XB)^T = I$$

donde X es una matriz simétrica ($X^T = X$) y se sabe que la matriz $A + B^T$ es invertible.

$$(A + B^T)^{-1}$$

a) Usando el álgebra de matrices, demuestre que $X = (A + B^T)^{-1}$.

$$X \cdot A = B \quad \text{, } X \text{ tiene inversa}$$

$$A = B \cdot X^{-1}$$

$$AX + (X \cdot B)^T = I$$

$$AX + B^T \cdot X^T = I$$

$$AX + B^T X = I$$

$$(A + B^T)X = I$$

$$X = I(A + B^T)^{-1}$$

$$\boxed{X = (A + B^T)^{-1}}$$

$$(A \cdot B)^T = B^T \cdot A^T$$

$$X^T = X$$

$$AX + BX = (A + B)X$$

$$(A + B^T) = (A + B^T)^{-1}$$

$$AI = IA = A$$

b) Calcule en forma explícita la matriz X si

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$$

$$X = (A + B^T)^{-1}$$

$$B^T = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} + \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

$$A + B^T = \begin{pmatrix} 3 & 0 \\ 2 & 4 \end{pmatrix}$$

$$(A + B^T)^{-1} = \left(\begin{array}{cc|cc} 3 & 0 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{cc|cc} 3 & 0 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array} \right)$$

$$\frac{1}{3} \cdot \widetilde{F1} \quad \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{3} & 0 \\ 2 & 4 & 0 & 1 \end{array} \right)$$

$$-2 \cdot \widetilde{F1} + \widetilde{F2} \quad \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{3} & 0 \\ 0 & 4 & -\frac{2}{3} & 1 \end{array} \right)$$

$$\frac{1}{4} \cdot \widetilde{F2} \quad \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{3} & 0 \\ 0 & 1 & -\frac{1}{6} & \frac{1}{4} \end{array} \right)$$

$$(A + B^T)^{-1} = \begin{pmatrix} \frac{1}{3} & 0 \\ -\frac{1}{6} & \frac{1}{4} \end{pmatrix}$$

#7 Resuelva el siguiente sistema de ecuaciones por el método de Gauss - Jordan

(5 Pts)

$$\begin{cases} x + 2y + w = 10 \\ x + 2y + z + w = 5 \\ -2y + 2z + 2w = 4 \end{cases}$$

$$\begin{pmatrix} 1 & 2 & 0 & 1 & | & 10 \\ 1 & 2 & 1 & 1 & | & 5 \\ 0 & -2 & 2 & 2 & | & 4 \end{pmatrix}$$

$$-F_1 + \widetilde{F}_2 \begin{pmatrix} 1 & 2 & 0 & 1 & | & 10 \\ 0 & 0 & 1 & 0 & | & -5 \\ 0 & -2 & 2 & 2 & | & 4 \end{pmatrix}$$

$$F_2 \leftrightarrow F_3 \begin{pmatrix} 1 & 2 & 0 & 1 & | & 10 \\ 0 & -2 & 2 & 2 & | & 4 \\ 0 & 0 & 1 & 0 & | & -5 \end{pmatrix}$$

$$-\frac{1}{2} \cdot \widetilde{F}_2 \begin{pmatrix} 1 & 2 & 0 & 1 & | & 10 \\ 0 & 1 & -1 & -1 & | & -2 \\ 0 & 0 & 1 & 0 & | & -5 \end{pmatrix}$$

$$-2 \cdot F_2 + \widetilde{F}_1 \begin{pmatrix} 1 & 0 & 2 & 3 & | & 14 \\ 0 & 1 & -1 & -1 & | & -2 \\ 0 & 0 & 1 & 0 & | & -5 \end{pmatrix}$$

$$\begin{matrix} x & y & z & w \\ -2 \cdot F_3 + \widetilde{F}_1 \\ F_3 + \widetilde{F}_2 \end{matrix} \begin{pmatrix} 1 & 0 & 0 & 3 & | & 24 \\ 0 & 1 & 0 & -1 & | & -7 \\ 0 & 0 & 1 & 0 & | & -5 \end{pmatrix}$$

$$x + 3w = 24 \rightarrow x = 24 - 3w$$

$$y - w = -7 \quad y = w - 7$$

$$z = -5 \quad z = -5$$

$$\boxed{\text{Sol } (24 - 3w, w - 7, -5, w)}$$